

Ephaptic Quantum Energy Teleportation and Quantum Ephaptic Coupling: A Unified Mechanism for Energy–Charge Recombination Across Lipid Barriers

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Abstract

Two distinct quantum mechanisms have recently been proposed for lateral, non-synaptic interactions between adjacent ion channel pores embedded in lipid membranes. The first, ephaptic quantum energy teleportation (EQET), enables the transfer of energy—but not particles—between pores via correlations in a shared electromagnetic field. The second, quantum ephaptic coupling (QEC), permits low-probability lateral tunneling of ions through the intervening lipid matrix, thereby transferring charge but not directed energy. In this work we unify these two mechanisms into a composite process in which energy and charge, transported by distinct physical channels, recombine at a target pore.

A key refinement introduced here is the promotion of ionic mass from a fixed scalar to a quantum operator, yielding mass-conditioned tunneling amplitudes and energy-extraction probabilities. This operator treatment reveals asymmetries in transport: heavier effective-mass components suppress tunneling yet enhance local energy capture. Furthermore, by extending the perturbative expansion to third order in the Dyson series, we identify the precise interaction terms that enable recombination, coupling mass, charge, and field degrees of freedom into a coherent process.

The resulting framework provides a microscopic recombination probability that quantifies the likelihood that a tunneled ion becomes the carrier of teleported energy. This mechanism is juxtaposed with well-established quantum effects in photosynthesis, highlighting a shared structural motif: the separation of energy propagation from charge motion followed by localized recombination. Limitations, scaling behavior, and physical constraints are discussed explicitly. No claims are made beyond the internally consistent framework developed here.

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1 Introduction

Biological membranes host a dense population of ion channels whose collective behavior underlies electrical signaling in neurons, muscle fibers, and plant cells. Traditionally, interactions between nearby channels have been described either through direct ionic conduction (synapses, gap junctions) or through classical ephaptic coupling mediated by extracellular electric fields. More recently, two quantum-inspired extensions of ephaptic interaction have been proposed.

The first, which we call *quantum ephaptic coupling* (QEC), models adjacent ion channel pores as quantum subsystems weakly coupled through lateral tunneling of ions across the lipid bilayer separating them. The lipid region acts as a high, thin potential barrier in the membrane plane, allowing exponentially suppressed but nonzero tunneling amplitudes. QEC therefore enables charge transfer between pores without classical conduction pathways.

The second, *ephaptic quantum energy teleportation* (EQET), adapts the framework of quantum energy teleportation (QET) to membrane systems. In this picture, an ion in one pore acts as a local probe of the joint electromagnetic field spanning multiple pores and the intervening lipid. A local interaction injects energy into the field at one pore, while pre-existing field correlations allow energy extraction at a distant pore without any particle traversing the barrier.

Individually, QEC and EQET transfer different physical quantities: QEC transfers charge, while EQET transfers energy. In this document we develop a unified framework in which these two processes operate concurrently, allowing energy and charge to recombine at a target pore. The arriving charge carrier then becomes the bearer of the teleported energy. This merged mechanism respects locality, causality, and energy conservation, while exhibiting non-classical features reminiscent of quantum photosynthesis.

To explore this unified framework in concrete terms, we now turn to the physical geometry of ion channels and the membrane environment, establishing the spatial and material context in which QEC and EQET can operate

2 Physical Geometry and System Description

We consider two ion channel pores, labeled 1 and 2, embedded in a lipid membrane and separated laterally by a distance d in the membrane plane. Each pore supports longitudinal ionic currents along its axis, normal to the membrane. The lipid matrix between pores is electrically insulating, hydrated, and noisy, but thin enough to permit weak quantum tunneling.

The system is decomposed into three subsystems:

1. The ionic degrees of freedom in pore 1.
2. The ionic degrees of freedom in pore 2.
3. The joint electromagnetic field and polarization field spanning pore 1, the lipid divider, and pore 2.

Crucially, the electromagnetic field is not separable into independent pore-local fields; its ground or near-equilibrium state contains spatial correlations across the lipid region. These correlations are the resource exploited by EQET.

Having defined the structural subsystems, we next refine the microscopic description by treating ionic mass not as a fixed scalar but as a quantum operator, thereby uncovering subtler dependencies in tunneling and energy extraction.

3 Quantum Treatment of Ionic Mass

In the preceding sections, the ionic mass was treated as a fixed parameter. We now extend the framework by adopting the formulation developed in [1], wherein mass is promoted to a quantum observable. This refinement is essential in regimes where confinement, tunneling, and energy extraction occur on comparable time scales.

3.1 Mass as an Operator

We replace the scalar mass m by a Hermitian operator $\hat{m}$ acting on an auxiliary mass Hilbert space,

$$m \rightarrow \hat{m}, \tag{1}$$

with nontrivial commutation relations

$$[\hat{m}, \hat{x}] \neq 0, \quad [\hat{m}, \hat{p}] \neq 0, \quad (2)$$

as derived in [1]. The ionic kinetic energy operator therefore becomes

$$\hat{H}_{\text{kin}} = \frac{\hat{p}^2}{2\hat{m}}, \quad (3)$$

and the unperturbed Hamiltonian for each pore is written as

$$\hat{H}_0 = \frac{\hat{p}^2}{2\hat{m}} + V_{\text{well}}(\hat{x}). \quad (4)$$

The eigenstates of $\hat{H}_0$ are no longer simple sine functions, but acquire mass-dependent corrections. Accordingly, ionic motion, tunneling rates, and characteristic frequencies become conditional on the mass state.

3.2 Implications for Tunneling

The charge-exchange (QEC) Hamiltonian retains its form,

$$\hat{H}_{\text{QEC}} = J \left(\hat{c}_1^\dagger \hat{c}_2 + \hat{c}_2^\dagger \hat{c}_1 \right), \quad (5)$$

but the tunneling coefficient becomes an operator-valued quantity,

$$J \rightarrow \hat{J}(\hat{m}), \quad (6)$$

with semiclassical scaling

$$\hat{J}(\hat{m}) \sim \exp \left[- \int dx \sqrt{\frac{2\hat{m}}{\hbar^2} (V - E)} \right]. \quad (7)$$

As a result, continuous tunneling entangles the ionic position with its mass state, leading to mass-conditioned charge delocalization already at first order in degenerate perturbation theory.

3.3 Second-Order Wavefunction Structure

To second order in time-dependent perturbation theory, the total wavefunction takes the form

$$|\Psi(t)\rangle = |\Psi_{\text{ion}}^{(1)}\rangle \otimes |0\rangle_{\text{field}} + \sum_{\lambda,\mu} C_{\lambda\mu}(t) |\psi_{\mu}\rangle_{\text{ion}} \otimes |1_{\lambda}\rangle_{\text{field}}, \quad (8)$$

where μ labels mass eigenstates and λ field modes. Energy extraction via EQET therefore depends explicitly on the ionic mass sector, even though no energy is yet carried by the ion itself.

3.4 On the Appearance of Third-Order Effects

Within Sakurai's framework, recombination and energy capture by the ion require one additional interaction beyond second order. At third order in the Dyson expansion, matrix elements of the form

$$\langle f | V_{\text{rec}} V_{\text{EQET}} V_{\text{QEC}} | i \rangle \quad (9)$$

become nonzero, allowing the field-excited intermediate states to project back onto ionic degrees of freedom.

Crucially, when mass is treated as an operator, third-order perturbation theory produces *mass-selective recombination amplitudes*. Heavier effective-mass components experience suppressed tunneling but enhanced local energy capture, yielding an intrinsic asymmetry in transport and recombination probabilities. These effects arise without violating unitarity or detailed balance, and reflect the composite structure of the ion–field–mass Hilbert space.

3.5 Summary

Promoting ionic mass to a quantum observable modifies tunneling, velocity, and energy-extraction processes in a controlled and perturbatively consistent manner. This sets the stage for a focused treatment of quantum ephaptic coupling (QEC), where tunneling across lipid barriers becomes the primary mechanism of charge transfer.

4 Quantum Ephaptic Coupling (QEC)

We first summarize QEC in its minimal form. Let $\psi_1(x)$ and $\psi_2(x)$ denote effective lateral wavefunctions associated with ions localized in pore 1 and pore 2, respectively. After integrating out the lipid degrees of freedom, the effective Hamiltonian takes the form

$$H_{\text{QEC}} = H_1 + H_2 + J_{\text{lipid}} (|1\rangle \langle 2| + |2\rangle \langle 1|), \quad (10)$$

where J_{lipid} is a tunneling matrix element determined by the lipid barrier.

For a rectangular lateral barrier of height V_L and width d , the tunneling amplitude scales as

$$J_{\text{lipid}} \propto e^{-\kappa d}, \quad \kappa = \sqrt{\frac{2m^*(V_L - E_{\perp})}{\hbar^2}}, \quad (11)$$

where m^* is an effective ionic mass and $E_{\perp}$ is the lateral kinetic energy.

The associated tunneling probability is

$$P_{\text{tun}} \sim e^{-2\kappa d}. \quad (12)$$

QEC therefore provides a mechanism for rare but genuine lateral transfer of charge between adjacent pores, without invoking pore gating or extracellular conduction.

While QEC highlights the rare but genuine lateral transfer of charge, it leaves open the question of how energy itself might propagate without particle motion. This motivates the introduction of EQET, which addresses energy transfer through correlated fields.

5 Ephaptic Quantum Energy Teleportation (EQET)

EQET adapts the logic of quantum energy teleportation to membrane systems. An ion in pore 1 is treated as a probe that locally couples to the joint electromagnetic field. The interaction Hamiltonian is written as

$$H_{\text{int}}(t) = g(t) \hat{q}_1 \hat{G}, \quad (13)$$

where $\hat{q}_1$ is the charge operator of the probe ion and

$$\hat{G} = \int d^3x a_1(\mathbf{x}) \hat{E}(\mathbf{x}) \quad (14)$$

is a field operator localized near pore 1.

A time-dependent coupling $g(t)$ injects energy into the field locally at pore 1. Because the field state contains correlations extending to pore 2, a local operation or passive interaction at pore 2 can extract energy. Importantly:

- No ion traverses the lipid.
- No classical energy current flows through the divider.
- Energy appears at pore 2 as a change in local field or polarization energy.

Energy conservation is maintained globally: positive energy injected at pore 1 is balanced by negative-energy density fluctuations and subsequent extraction at pore 2.

With QEC and EQET defined as distinct channels—one for charge, the other for energy—we are now prepared to examine their complementarity and the possibility of their concurrent operation.

6 Complementarity of QEC and EQET

QEC and EQET act on distinct physical channels:

- QEC transfers *charge* but does not guarantee directed energy transfer.
- EQET transfers *energy* but not particles.

This complementarity suggests a natural unification. Suppose the probe ion in pore 1 both:

1. interacts locally with the joint field, triggering EQET and depositing energy at pore 2;
2. tunnels laterally through the lipid barrier with low probability, arriving physically in pore 2.

The two processes are independent but can occur on comparable timescales. When both occur, pore 2 temporarily hosts:

- a localized excess of field or polarization energy;
- an additional ionic charge.

The central question is whether these two can *recombine*, such that the arriving ion becomes the carrier of the teleported energy.

The coexistence of charge tunneling and energy teleportation naturally raises the question of recombination: under what conditions can an ion arriving via QEC absorb energy deposited by EQET? We formalize this process next.

7 Definition of Recombination

We define recombination as the process by which a tunneled ion absorbs energy deposited locally in pore 2 by EQET, converting field or polarization energy into ionic kinetic or configurational energy.

Three probabilities enter:

1. $P_{\text{QET}} \approx 1$, conditional on executing the EQET protocol.
2. $P_{\text{tun}} \sim e^{-2\kappa d}$, the tunneling probability.
3. P_{rec} , the recombination probability.

The overall probability that a tunneled ion carries teleported energy is

$$P_{\text{carrier}} = P_{\text{tun}} P_{\text{rec}}. \quad (15)$$

To move beyond abstract probabilities, we must identify the microscopic interactions that enable recombination. Section 8 develops this mechanism in detail, focusing on Coulomb coupling between tunneled ions and localized field excitations.

8 Microscopic Mechanism for Recombination

At pore 2, EQET produces a localized excited field state $|\epsilon\rangle$ with finite lifetime τ_{field} . The arriving ion has a wavefunction $\psi_{\text{ion}}(\mathbf{x}, t)$ localized within the pore.

The dominant interaction enabling recombination is Coulomb coupling between the ion and the local electric field:

$$H_{\text{rec}} = q \hat{\mathbf{x}} \cdot \hat{\mathbf{E}}_{\text{loc}}. \quad (16)$$

This interaction allows transitions of the form

$$\left| \psi_{\text{ion}}^{(0)} \right\rangle \otimes |\epsilon\rangle \longrightarrow \left| \psi_{\text{ion}}^{(1)} \right\rangle \otimes |0\rangle, \quad (17)$$

where the ion absorbs the field energy.

Using Fermi's Golden Rule, the recombination probability scales as

$$P_{\text{rec}} \propto \left| \left\langle \psi_{\text{ion}}^{(1)} \right| H_{\text{rec}} \left| \psi_{\text{ion}}^{(0)} \right\rangle \right|^2 \tau_{\text{field}}. \quad (18)$$

Estimating matrix elements yields

$$P_{\text{rec}} \sim \left(\frac{q E_{\text{loc}} \ell}{\Delta E} \right)^2 \Theta(\tau_{\text{ion}} < \tau_{\text{field}}), \quad (19)$$

where ℓ is the pore localization length and ΔE characterizes ionic level spacing or thermal broadening.

Having established the microscopic pathway for recombination, we now integrate tunneling and recombination probabilities into a single scaling law that quantifies the likelihood of energy-charge carriers emerging in pore 2.

9 Combined Scaling Law

Combining tunneling and recombination,

$$P_{\text{carrier}} \sim e^{-2\kappa d} \left(\frac{q E_{\text{QET}} \ell}{\Delta E} \right)^2. \quad (20)$$

Thus:

- Arrival of a charge carrier is exponentially rare.
- Conditional on arrival, energy capture can be efficient.

This mirrors other biological quantum processes where rare events have disproportionate functional impact.

Although the combined scaling law captures the essential rarity and conditional efficiency of the process, real biological membranes introduce geometric imperfections. Section 10 examines how sub-pore misalignment modulates recombination outcomes.

10 Geometric Misalignment as a Modulator of Recombination

Section 7 established a microscopic mechanism by which a tunneled ion can absorb locally deposited EQET energy via Coulomb coupling to an excited field mode at pore 2. That analysis assumed an idealized pore geometry with implicit spatial symmetry. We now relax this assumption by introducing geometric misalignment at the sub-pore level and show that it naturally enhances recombination without altering the underlying tunneling physics. This refinement serves as a bridge between the microscopic interaction Hamiltonian and the macroscopic scaling law derived in the next section.

10.1 Sub-pore misalignment of ion channels

In realistic lipid membranes, ion channels are not arranged in perfectly registered configurations. Individual channel proteins are laterally staggered with respect to one another on the membrane surface, producing a characteristic offset $\Delta\mathbf{x}$ in the membrane plane. This geometric misalignment is the membrane-scale analogue of nodal misalignment in ephaptically coupled axon tracts, where changes in nodal placement modify local propagation dynamics without altering inter-axonal coupling strengths.

At the pore level, such misalignment reshapes the spatial structure of the local electric and polarization fields. Accordingly, the excited field state generated by EQET near pore 2 is no longer centrally symmetric but instead exhibits spatially displaced maxima and minima tied to $\Delta\mathbf{x}$.

10.2 Effect on EQET-localized energy

Because EQET operates through pre-existing field correlations rather than particle transport, sub-pore misalignment does not suppress energy teleportation. Instead, it redistributes teleported energy into spatially structured polarization modes near pore 2. In particular, misalignment reduces destructive interference between overlapping field modes associated with neighboring channel mouths.

As a result, the effective local energy density available for capture becomes a function of channel geometry,

$$E_{\text{loc}} \rightarrow E_{\text{loc}}(\Delta \mathbf{x}), \quad (21)$$

with $E_{\text{loc}}(\Delta \mathbf{x})$ generically enhanced for offsets comparable to the pore localization length.

10.3 Misalignment-enhanced energy recapture after tunneling

When a rare QEC tunneling event delivers an ion into pore 2, the ion encounters this spatially structured excited field rather than a uniform background. The recombination Hamiltonian introduced in Section 7 therefore acquires an explicit geometric dependence,

$$H_{\text{rec}}(\Delta \mathbf{x}) = q \hat{\mathbf{r}} \cdot \hat{\mathbf{E}}_{\text{loc}}(\Delta \mathbf{x}). \quad (22)$$

This dependence has an important dynamical consequence. As in misaligned ephaptically coupled axons—where saltatory energy localizes preferentially at specific nodes—channel misalignment biases the interaction toward energy capture rather than re-emission. Local detailed balance is broken in favor of recombination, even though global energy conservation remains intact.

Applying Fermi's Golden Rule yields a recombination probability of the form

$$P_{\text{rec}} \sim \left(\frac{q E_{\text{loc}}(\Delta \mathbf{x}) \ell}{\Delta E} \right)^2 \Theta(\tau_{\text{ion}} < \tau_{\text{field}}), \quad (23)$$

indicating that misalignment enhances recombination efficiency without modifying the exponentially small tunneling probability.

10.4 Transition to the global scaling law

This geometric refinement completes the microscopic picture required to interpret the combined energy–charge transfer process. Tunneling remains the rate-limiting step, while recombination efficiency is controlled by local field structure and channel geometry. In the following section, these elements are combined into a single scaling law that quantifies the probability that a tunneled ion becomes the carrier of teleported energy.

With geometric refinements incorporated, the mechanism acquires a structural resemblance to other biological quantum processes. This invites a direct comparison with quantum photosynthesis, where energy

and charge are likewise separated and recombined.

11 Juxtaposition with Quantum Photosynthesis

Quantum photosynthesis, as studied by Whaley, Lidar, Engel, and others, exhibits a strikingly similar structural motif. In light-harvesting complexes:

- Energy propagates as excitons without charge motion.
- The protein environment assists coherence.
- Energy is trapped at a reaction center, where charge separation occurs.

In the EQET–QEC mechanism:

- Energy propagates via field correlations without ion motion.
- The lipid–water environment sustains correlations.
- Energy recombines with charge at a target pore.

Both systems separate energy transport from charge transport, then reunite them locally. The difference lies in implementation: excitonic hopping versus energy teleportation and tunneling.

To ground this analogy, we consider the broader biological setting of leaves, where ionically active membranes and lipid barriers provide a natural arena for such processes—even if not yet experimentally confirmed.

12 Placement in the Leaf

Leaves provide a natural arena for this juxtaposition. Beyond chloroplasts, leaf tissue contains:

- Ionically active membranes (guard cells, vacuoles).
- Lipid barriers and hydrated environments.
- Electrical signaling mediated by ions.

No claim is made that plants exploit EQET–QEC. Rather, the leaf demonstrates that biological matter can support quantum coherence, correlated environments, and energy–charge recombination in warm, noisy conditions.

Yet before drawing speculative conclusions, it is essential to acknowledge the limitations of the framework, particularly the small tunneling probabilities and decoherence effects that constrain its biological plausibility.

13 Limitations

Several limitations must be emphasized:

1. Tunneling probabilities are extremely small.
2. Decoherence limits field correlation lifetimes.
3. Thermal noise competes with small energy gains.
4. No direct experimental evidence currently exists for EQET–QEC in biological membranes.

The mechanism is therefore not proposed as a dominant transport pathway, but as a possible microscopic correction or modulation channel.

The mechanism is therefore not proposed as a dominant transport pathway, but as a possible microscopic correction or modulation channel. In light of these constraints, the conclusion synthesizes the theoretical consistency of EQET–QEC while emphasizing the need for further scrutiny.

14 Conclusion

By unifying ephaptic quantum energy teleportation with quantum ephaptic coupling, we have constructed a logically consistent mechanism in which energy and charge are transported by distinct quantum processes and recombine at a target pore. The resulting framework respects physical constraints and mirrors, at a structural level, the well-established separation and recombination of energy and charge in quantum photosynthesis. Whether nature exploits such a mechanism remains open, but its internal consistency warrants further theoretical and experimental scrutiny.

References

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